

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-4; module 5 through memory operations. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) How many cells are in a four-variable Karnaugh map? (1)
(A) 4
(B) 8
(C) 16
(D) 32
- 2) How many input variables differ between two adjacent Karnaugh-map cells? (1)
(A) 1
(B) 2
(C) 3
(D) 4
- 3) Which is a valid group size when minimizing a sum-of-products expression on a Karnaugh map? (1)
(A) 3
(B) 5
(C) 6
(D) 8
- 4) For $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$, with A most significant, which expression is equivalent? (1)
(A) $B \oplus D$
(B) $\neg B \neg D + BD$
(C) $B + D$
(D) BD
- 5) How can a two-input NAND gate act as an inverter for input X? (1)
(A) Tie both inputs to 0
(B) Leave one input open
(C) Tie both inputs to X
(D) Tie one input to 0

Section B: Theory questions (25 marks)

- 6) Using 5-bit two's complement, evaluate (i) $11 + 7$, (ii) $-9 + 5$, and (iii) $6 - 13$. Give the binary result, its signed interpretation and whether signed overflow occurs in each case. (3)
- 7) Implement $F(A, B, C) = A \oplus B \oplus C$ using a 4-to-1 multiplexer. Choose A and B as select inputs and specify I_0, I_1, I_2, I_3 ; an inverter is available. (2)
- 8) Design a synchronous two-bit counter with T flip-flops and enable input E. For $E = 1$, the state sequence is $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$; for $E = 0$, the state holds. Draw the transition table, derive both T-input equations and draw the circuit. (5)
- 9) Write synthesizable Verilog for a four-bit ring counter with positive-edge clock and active-high synchronous reset. Reset must load 0001; subsequent clocks rotate the one toward the next more significant bit, wrapping from bit 3 to bit 0. (3)
- 10) Explain why an asynchronous ripple counter can briefly produce incorrect decoded outputs. Contrast its clock connection with that of a synchronous counter. (2)
- 11) Design a two-bit unsigned magnitude comparator for $A = a_1a_0$ and $B = b_1b_0$. Derive Boolean expressions for $A > B$, $A = B$ and $A < B$. Draw a gate-level diagram and verify the outputs for $A = 10_2$, $B = 01_2$. (5)
- 12) A byte-addressable memory stores 32-bit words. How many words fit in 4 KiB? How many address bits select individual bytes? If a word starts at hexadecimal address $0x1000$, list the byte addresses it occupies and the starting address of the next word. (3)
- 13) Describe a memory-read operation using the memory address register (MAR), memory data register (MDR) and read control signal. State what changes for a memory-write operation. (2)

End of question paper